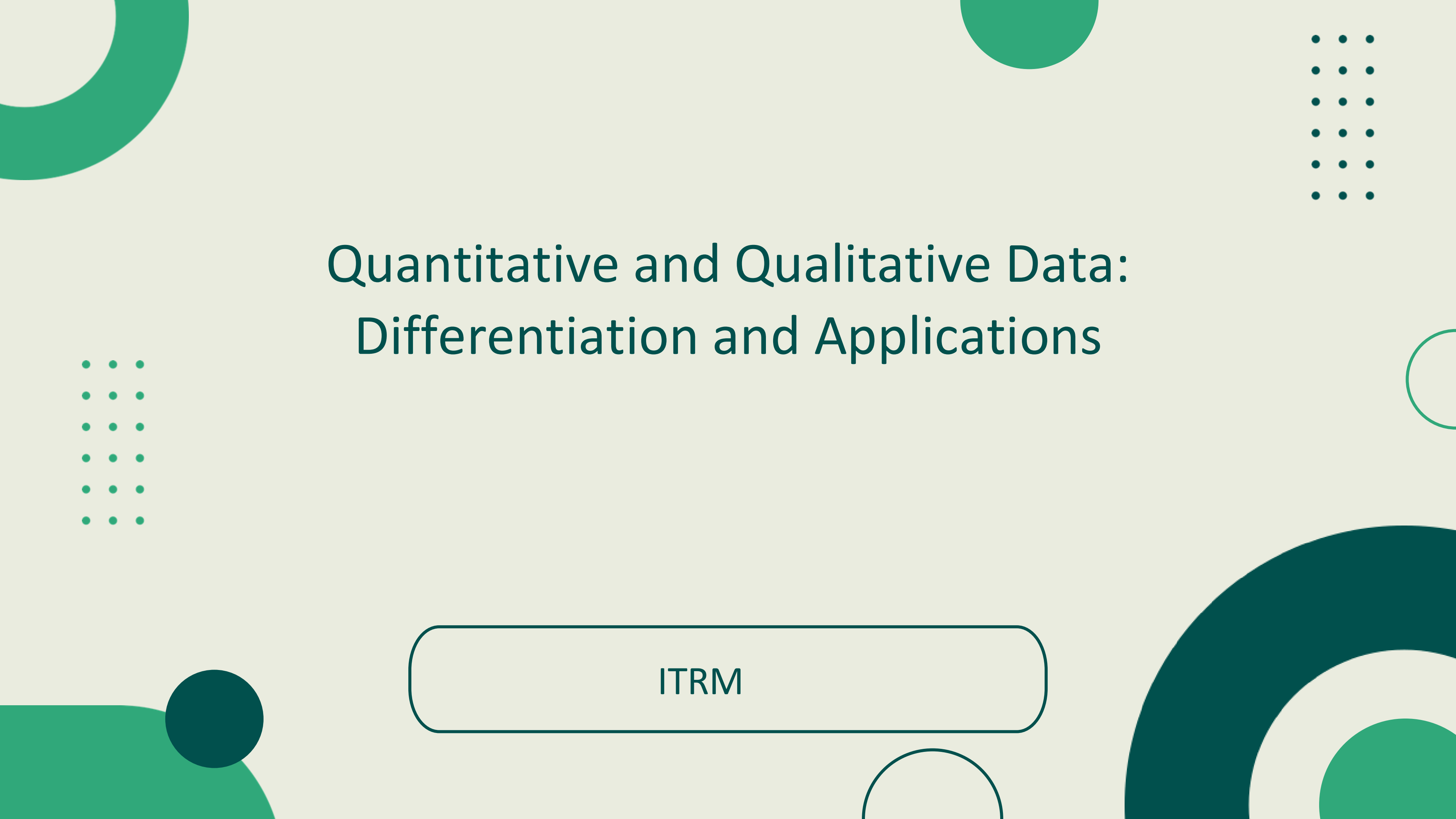




Quantitative and Qualitative Data: Differentiation and Applications

ITRM

Overview of Data Types in Research

Quantitative Data: Involves numerical measurements and statistical analysis, enabling precise, reproducible findings across large samples.

Qualitative Data: Captures descriptive information, and narratives, providing contextual depth and interpretive insight.

Both data types are essential — **quantitative methods** reveal patterns and trends across nations, while **qualitative approaches** uncover the complex social, political, and cultural dynamics that numbers alone cannot explain.

Key Distinctions

Quantitative Data

Numerical & measurable

Focus: How much/many/often

Statistical analysis, Large sample sizes,

Predetermined categories

Example: Population = 115 Million



Key Distinctions

Qualitative Data

Descriptive & interpretive

Focus: Why/How

Smaller purposive samples

Emergent categories

Example: "People migrate because of better job opportunities."

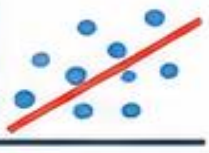




Characteristics of Quantitative Data



• Key Properties •

1		Numerical Representation	Data is expressed in numbers, allowing precision and measurement.	Example: Temperature: 28°C Population: 115 million	 
2		Objectivity and Standardization	Collected using consistent methods, reducing bias and personal interpretation.	Example: Using the same survey questions for all respondents.	
3		Reproducibility and Verifiability	Results can be repeated by other researchers and verified.	Example: Another researcher gets the same results using the same data.	
4		Aggregation for Generalization	Data from many individuals or cases can be combined to draw conclusions for a large population.	Example: Surveying 1,000 students to represent all students in a university.	
5		Statistical Manipulation Capabilities	Allows the use of statistical techniques to analyze and understand data.	Example: Using averages, percentages, regression, or correlation.	
6		Relationship Analysis	Helps identify relationships and patterns between variables.	Example: Studying the relationship between study hours and exam scores.	
7		Supports Evidence-Based Policy and Comparative Analysis	Provides reliable evidence for decision-making and comparing groups or countries.	Example: Comparing GDP growth rates of different countries to guide policy.	



KEY TAKEAWAY

Quantitative data uses numbers to produce accurate, reliable, and generalizable results that support informed decisions.

In short:



Collect



Analyze



Compare



Decide



Quantitative Data: Structural Features



Quantitative data is numerical, structured, and analyzed using statistical methods.

1		High Dimensionality	Many variables are measured at the same time across large datasets.	Example: A health survey collects data on age, income, education, weight, blood pressure, and lifestyle for thousands of people.
2		Structured Data	Data is organized in a fixed format such as matrices, databases, and spreadsheets.	Example: Data stored in Excel sheets or SQL databases with rows (cases) and columns (variables).
3		Measurement Scales	Data is measured using different levels of scales.	Types: Nominal Categories with no order (e.g., gender) Ordinal Categories with order (e.g., ranking) Interval Equal intervals, no true zero (e.g., °C) Ratio Equal intervals, true zero (e.g., weight)
4		Distribution Analysis	Examines the shape and spread of data.	Example: Checking if data is normally distributed, skewed left or right, or has high/low variance.
5		Relationship Analysis	Studies relationships between variables.	Example: Using correlation to see if study hours and exam scores are related.
6		Advanced Modeling Techniques	Uses statistical models to predict or explain outcomes.	Example: Regression analysis or multivariate modeling to predict factors that affect economic growth.



KEY TAKEAWAY

Quantitative data is numerical, organized, measured at different scales, and analyzed to find patterns, relationships, and make data-driven decisions.

EXAMPLES IN REAL LIFE



Population Statistics



Economic Indicators



Health Data



Election Results



THINK-PAIR-SHARE

Give one example of quantitative data in international studies.

Share your answers!



Qualitative Data: Key Attributes & Features

Exploring meanings, experiences, and perspectives behind human behavior.



1. Rich Contextual and Narrative Depth

Provides detailed descriptions of contexts, settings, and experiences.

► *Example: Personal stories from interviews about migration experiences.*



2. Flexibility and Adaptability in Data Collection

Methods can be adjusted as the study progresses to explore new insights.

► *Example: Changing interview questions based on participants' responses.*



3. Subjective Interpretation and Researcher Reflexivity

Researchers interpret data while being aware of their own biases and perspectives.

► *Example: Reflecting on how your experiences shape your interpretation.*



4. Holistic Perspective Capturing Social Complexity

Looks at the whole picture and understands how different factors are connected.

► *Example: Understanding how culture, family, and community influence decisions.*



5. Narrative-Based Documentation of Lived Experiences

Focuses on real-life stories and meanings shared by participants.

► *Example: Life stories, diaries, focus group discussions, field notes.*



QUALITATIVE DATA: DISTINCTIVE FEATURES



Depth over Breadth

Focuses on in-depth understanding of a few cases rather than broad generalization. Reveals nuanced meanings and complex social phenomena that numbers cannot capture.



Accommodates Ambiguity and Contradictions

Embraces complexity and inconsistency as meaningful data. Iterative data collection and analysis allow continuous refinement, with emerging insights shaping subsequent inquiry cycles.



KEY TAKEAWAY

Qualitative data helps us understand the “why” and “how” behind human experiences. It gives voice to people and captures the richness of real-life stories and social contexts.



THINK-PAIR-SHARE

Give one example of qualitative data you might collect in a study about students' experiences in online learning.

Quantitative Data Collection Methods – Primary Techniques



Surveys and questionnaires: Structured instruments capturing numerical responses from large samples

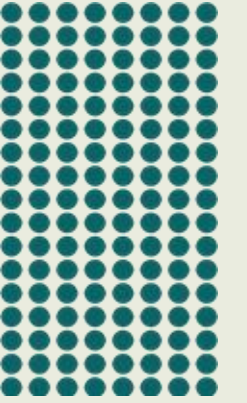
Controlled experiments: Manipulating variables under standardized conditions to measure outcomes

Structured observations: Systematic recording of behaviors or events using predefined categories

Administrative data extraction: Drawing numerical records from institutional or governmental databases

Population censuses: Comprehensive enumeration of entire populations providing large-scale demographic data

Quantitative Collection: Technology-Enabled Approaches



Web Analytics: Captures millions of behavioral data points from online platforms and user interactions

Sensor Networks: Continuous real-time data streams from environmental, infrastructure, and IoT sensors

Automated Systems: Large-scale structured databases generated through automated recording and processing

API Data Feeds: Real-time access to structured datasets from social media, financial, and governmental platforms

Satellite Imagery: High-resolution geospatial grids enabling remote sensing and large-scale geographic analysis

Qualitative Data Collection Methods – Traditional Approaches

In-depth interviews: one-on-one conversations exploring participants' perspectives, experiences, and meanings in rich detail

Focus groups: structured group discussions revealing shared attitudes, social norms, and collective understandings

Ethnographic observation: immersive fieldwork documenting behaviours, practices, and cultural contexts in natural settings

Document analysis: systematic review of texts, reports, policies, and records to identify themes and patterns

Case studies: in-depth investigation of a bounded unit (individual, group, event, or institution) to generate contextual insight



Qualitative Data Collection – Contemporary Approaches



Narrative interviews: capturing lived experiences and personal stories in depth

Digital ethnography: observing and analyzing online communities and virtual spaces

Visual methods: photography and video to document social phenomena
and participant realities

Participatory action research: co-creating knowledge with communities for
transformative outcomes





DATA SOURCES FOR QUANTITATIVE RESEARCH

Information comes from numbers, measurements, and structured data.

1

PRIMARY SOURCES



Data collected directly by the researcher from original observations or participants for the specific study.



Surveys & Questionnaires
(structured, close-ended questions)



Controlled Experiments
(manipulating variables and measuring results)



Structured Observations
(using checklists or rating scales)



Census Data & Field Measurements
(counts, measurements, readings)



Direct Administrative Data
(from records, forms, logs, or institutions)

2

SECONDARY SOURCES



Data already collected and compiled by others for different purposes and made available for public or research use.



Government Statistics
(PSA, NEDA, DOH, DepEd, etc.)



International Organizations
(World Bank, IMF, UN, WHO, OECD, etc.)



Academic Datasets & Publications
(journals, theses, research databases)



Commercial Databases
(Statista, Euromonitor, Nielsen, etc.)



NGO Reports & Documents
(surveys, reports, and studies)



Cross-National Comparative Datasets
(UN Data, PISA, World Development Indicators, etc.)



KEY TAKEAWAY

Quantitative research relies on reliable and structured data to produce accurate, measurable, and generalizable results.

GOOD DATA SOURCES → BETTER RESULTS



Collect



Organize



Analyze



Make Decisions



Quality data leads to informed decisions!



DATA SOURCES FOR QUALITATIVE RESEARCH

Information comes from **words**, **experiences**, and **real-life contexts**.



PRIMARY SOURCES

(Information collected directly from people)



Interviews – Asking individuals about their experiences, opinions, and perspectives.



Focus Group Discussions (FGD) – Group conversations to explore shared views and ideas.



Observations – Watching and recording behavior or events in their natural setting.



Personal Narratives – Stories and experiences shared by individuals.



Case Studies – In-depth investigation of a single individual, group, or event.



Diaries or Journals – Personal written records of thoughts, feelings, and events.



Photographs or Videos – Visual documents capturing people, places, or events.



SECONDARY SOURCES

(Information already collected by others)



Books – Written works providing background information and theories.



Journal Articles – Published research findings in academic journals.



Government Reports – Official reports from government agencies.



News Articles – Reports from newspapers, magazines, or online news.



Documentaries – Films or videos presenting real-life stories and issues.



Historical Documents – Old records, letters, photos, and other historical materials.



NGO Reports – Reports and publications from non-government organizations.



Social Media Posts – Public posts, comments, and discussions online.



Blogs and Online Forums – Online discussions and blogs on various topics.



EXAMPLE SCENARIO

Research Topic:

“Students’ Experiences with Online Learning”



Primary Sources

- Interviews with students
- Focus group discussion
- Classroom observation
- Student reflection journals



Secondary Sources

- Previous research articles
- DepEd or CHED reports
- News articles
- Educational blogs



QUICK ACTIVITY

Identify whether the source is **PRIMARY** or **SECONDARY**.

1



Interview with a tourist

2



PSA Report

3



Personal Diary

4



Journal Article

5



Focus Group Discussion

6



Documentary Film



KEY TAKEAWAY: Primary sources give first-hand information. Secondary sources provide existing information from others.



Collect



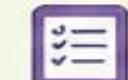
Organize



Analyze



Interpret



Make Decisions

Advantages of Quantitative Data



Research Benefits

Statistical generalization across populations · Comparative analysis across cases and regions · Objectivity and standardization in measurement · Strong support for evidence-based policy · Predictive modeling and forecasting capabilities

Practical Benefits

Efficiency in large-scale data collection · Replicability and verifiability of findings · Standardized protocols enabling consistency · Scalability to diverse contexts and populations · Integration with other datasets and systems

Advantages of Qualitative Data

Research Insights

Deep contextual understanding of phenomena; captures participant perspectives and lived experiences; reveals unexpected patterns and emergent themes; enables exploration of sensitive topics with nuance; supports theory development grounded in empirical observation.

Interpretative Depth

Explains the how and why behind social phenomena; accommodates ambiguity and contradictions within data; offers alternative explanations that quantitative methods may overlook; captures the complexity of human experience in international contexts.

Disadvantages of Quantitative Data



Methodological Limitations

Oversimplification of complex social phenomena; requires large sample sizes for statistical power; lacks nuance and depth; relies on predefined categories that may not capture reality; risk of measurement error distorting findings.

Contextual Constraints

Reductionism limits holistic understanding; response bias in surveys; missing data reduces validity; temporal lag in administrative datasets; cultural bias in standardized instruments across diverse contexts.

Disadvantages of Qualitative Data



Common Challenges

Subjectivity and researcher bias in interpretation • Limited generalizability beyond specific contexts • Time-consuming data collection and analysis • Difficulty in comparison across cases • Consistency and reliability issues across researchers

Practical Constraints

Requires sustained engagement and long-term fieldwork • Scalability limits: difficult to apply to large populations • Perceived lack of objectivity may reduce credibility • Replication challenges due to context-dependence

Data Management – Quantitative Systems

Storage

Relational databases · Data warehouses · Cloud storage solutions · Version control systems · Metadata management and documentation standards

Tools

Statistical software: R, STATA, SPSS · BI platforms: Tableau, Power BI · SQL databases · Programming: Python, Scala · Machine learning: TensorFlow

Data Management – Qualitative Systems

Organization

Coding frameworks for thematic categorization · QDA software integration · Document management and archiving · Annotation and memo writing · Audit trails for transparency and dependability

Software

NVivo: node-based coding, query functions, mixed methods support · Atlas.ti: network views, grounded theory workflows · MAXQDA: mixed methods, visual tools, team collaboration

Data Quality & Validity



Quantitative Standards: reliability and consistency of measures, internal and external validity, representativeness of samples, adequate response rates, and management of missing data.



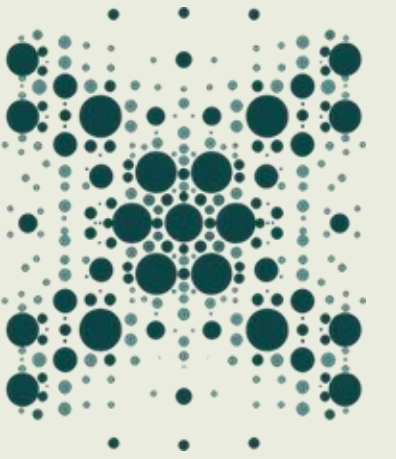
Qualitative Standards: credibility of findings, transferability to other contexts, dependability of the research process, confirmability of results, and authenticity of participant voices.

Integration and Mixed Methods Approaches

Quantitative data reveals prevalence and patterns; qualitative data explains mechanisms and meanings — together they provide comprehensive insight.

- Sequential: one method informs the other in phases
- Concurrent: both methods used simultaneously
- Explanatory: quantitative first, then qualitative to explain findings
- Exploratory: qualitative first, then quantitative to test findings
- Triangulation: cross-validate results from both approaches

Emerging Trends & Future Directions



Technological Innovations

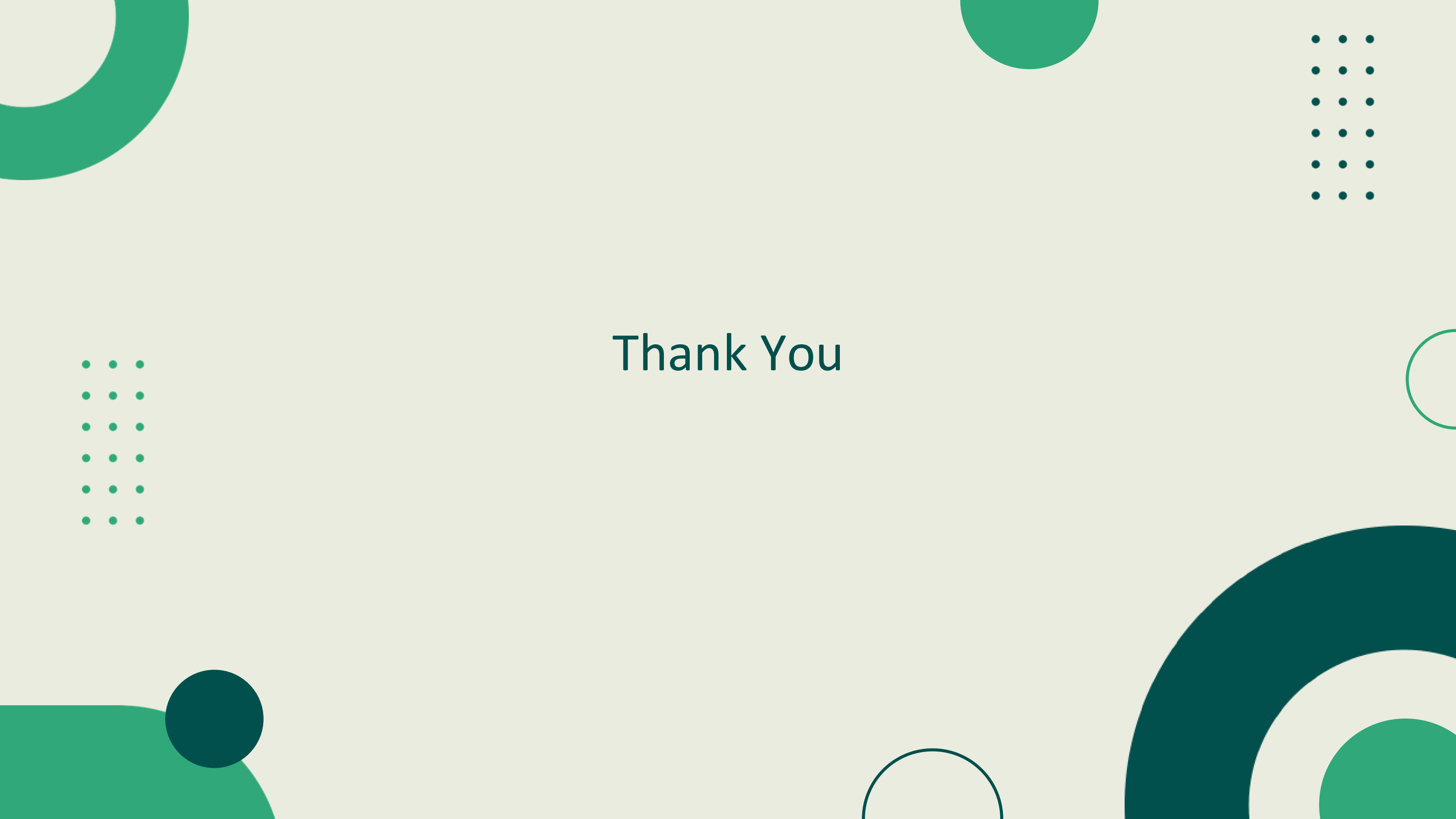
Big data analytics at scale · AI-powered text analysis and NLP · Geospatial technologies and GIS mapping · Blockchain for data transparency · IoT sensors for real-time data streams

Disciplinary Evolution

Data pluralism and methodological openness · Research ethics and data governance · Transparency and open science practices · Reflexivity in power dynamics and positionality

Summary & Key Takeaways

- Both quantitative and qualitative data are essential in international studies, offering complementary strengths for robust research.
- Quantitative data enables statistical generalization, precision, and large-scale comparison; qualitative data provides depth, context, and interpretive insight.
- Applications span international relations, comparative politics, and development studies, each benefiting from tailored data approaches.
- Effective data management — using appropriate software, coding frameworks, and quality standards — ensures research integrity.
- Emerging trends (big data, AI, mixed methods, ethical reflexivity) are reshaping how researchers collect, analyze, and interpret data.



Thank You